

SQL Business Insights Analysis

1. Project Overview

This project analyzes e-commerce sales data using SQL to generate business insights. The analysis helps understand customer behavior, product performance, revenue generation, and sales trends

2. Objectives

- Analyze sales performance
- Identify top-selling products
- Track customer spending
- Measure category-wise revenue
- Analyze monthly sales trends

3. Tools Used

- MySQL
- One Compiler
- GitHub

4. Database Tables

Customers

Stores customer information.

Products

Stores product details and prices.

Orders

Stores order information.

Order Details

Stores product-wise order details.

5. SQL Queries Performed

Query a): Total Revenue Analysis

```
SELECT SUM(p.Price * od.Quantity) AS TotalRevenue
FROM Products p JOIN OrderDetails od
ON p.ProductID = od.ProductID;
```

TotalRevenue
135000.00

Query b): Top Selling Products

```
SELECT p.ProductName, SUM(od.Quantity) AS TotalSold
FROM Products p
JOIN OrderDetails od
ON p.ProductID = od.ProductID
GROUP BY p.ProductName
ORDER BY TotalSold DESC;
```

ProductName	TotalSold
Laptop	2
HeadPhones	2
Shoes	2
Mobile	1
Watch	1

Query c): Revenue by Category

```
SELECT p.Category, SUM(p.Price * od.Quantity) AS Revenue
FROM Products p
JOIN OrderDetails od
ON p.ProductID = od.ProductID
GROUP BY p.Category;
```

Category	Revenue
Electronics	124000.00
Fashion	11000.00

Query d): Customer Spending Analysis

```
SELECT c.CustomerName, SUM(p.Price * od.Quantity) AS TotalSpent
FROM Customers c
JOIN Orders o
ON c.CustomerID = o.CustomerID
JOIN OrderDetails od
ON o.OrderID = od.OrderID
JOIN Products p
ON od.ProductID = p.ProductID
GROUP BY c.CustomerName
ORDER BY TotalSpent DESC;
```

CustomerName	TotalSpent
Rahul	55000.00
Neha	50000.00
Priya	26000.00
Amit	4000.00

Query e): Monthly Sales Trends

```
SELECT MONTH(o.OrderDate) AS Month,
SUM(p.Price * od.Quantity) AS Revenue
FROM Orders o
JOIN OrderDetails od
ON o.OrderID = od.OrderID
JOIN Products p
ON od.ProductID = p.ProductID
GROUP BY MONTH(o.OrderDate);
```

Month	Revenue
1	54000.00
2	31000.00
3	50000.00

6. Key Insights

- Electronics category generated the highest revenue.
- Laptop was among the top-selling products by revenue.
- Customer spending patterns were analyzed successfully.
- Monthly sales trends were identified using SQL queries.

7. Conclusion

This project demonstrates the use of SQL for business intelligence and data analysis. Various SQL operations such as joins, aggregations, and grouping were used to generate meaningful insights from e-commerce sales data.